

Mohib Amin

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WORK EXPERIENCE

SQL Developer Co-op | Environment Climate Change Canada

Jun. 2024 - May. 2025

- Re-engineered 15 high-traffic **PL/SQL** queries against a **100K+ record Oracle** database, cutting average execution time by **25%** through composite indexing and execution-plan rewrites.
- Built and enforced a role-based access control layer across 4 production schemas, eliminating 3 recurring data-integrity incidents flagged in quarterly audits **and reducing unauthorized access requests to zero.**
- Automated weekly schema-health reports using PL/SQL scripts, replacing a manual 2-hour review process and surfacing index bloat before it impacted downstream systems **or degraded report generation performance.**

Software QA Co-op | Environment Climate Change Canada

May. 2025 - Sept. 2025

- Authored and executed **50+ automated regression tests** in **Robot Framework**, catching 12 defects pre-release and lifting **automated coverage** from 45% to **75%** across 3 feature modules.
- Drove sprint-level defect triage in **Azure DevOps**, reducing average bug resolution time by 2 days by linking every user story to traceable acceptance criteria, **resulting in zero critical defects shipped to production**

EDUCATION

Bachelor of Engineering in Computer Engineering

Toronto Metropolitan University (Formerly known as Ryerson University)

Sept. 2021 - Apr. 2026

RELEVANT PROJECTS

Halal Stock Predictions — RL-Powered Investment Tool | Python, PPO (Reinforcement Learning), Flask, REST APIs

- Trained a **PPO reinforcement-learning** agent with **behavioral cloning** warm-start on **30+ equities**, outperforming a rule-based signal bot by **12%** in backtested returns over 6 months.
- Built a live dashboard (**Flask** + JavaScript) streaming real-time SPY/EMA signals, sector sentiment from news APIs, and **Kelly-criterion** position sizing on every page load.
- Implemented **RSI, MACD, EMA, and Bollinger Band** signal generators with confidence scoring, refreshing watchlist data every 5 minutes across 30+ tickers **and classifying each as BUY, HOLD, or SELL.**
- Engineered a **backtesting engine** comparing RL agent vs. rule-based bot across historical market conditions, visualizing portfolio value, drawdowns, and daily return distributions.

Smart Parking Pathfinder (Senior Design Project - 1st Place) | Arduino, Python (Flask, Socket.IO), JavaScript, HTML, CSS

- Architected a real-time smart parking solution connecting **Arduino**-based sensors, LED indicators, and servo-controlled gates to a live, browser-based parking map for continuous occupancy monitoring.
- Integrated a **Python** backend using **Flask** and **Socket.IO** to process serial data from hardware and broadcast real-time parking occupancy updates with **sub-second (200ms) latency** to connected clients.
- Constructed an interactive multi-level parking UI showcasing live availability, EV/accessibility classifications, and dynamic counts using responsive front-end components **with real-time WebSocket-driven state updates.**

AI-Powered NPC Behavior Modeling in Unity | Unity, C#, ML-Agents, Reinforcement Learning, PPO, Pytorch

- Built reinforcement learning NPCs in **Unity** using **ML-Agents** with custom observation/action spaces and reward design to enable adaptive navigation, combat decision-making, and context-aware behavioral responses within dynamic real-time gameplay environments, **replacing 500+ lines of traditional scripted behavior logic.**
- Improved agent convergence stability by **~30%** through iterative reward tuning and structured gameplay evaluation, outperforming traditional scripted NPC logic **in adaptive combat and navigation scenarios.**

SKILLS

Languages: Python, Java, JavaScript, SQL (PL/SQL), C, C++, C#, HTML/CSS, MATLAB, VHDL

AI/ML: PyTorch, Reinforcement Learning (PPO), Unity ML-Agents, NumPy, Scikit-learn, Pandas

Web & Data: Flask, FastAPI, Socket.IO, REST APIs, Oracle DB, Firebase, PostgreSQL, JSON, WebSockets, Playwright

Tools: Git, Linux/UNIX, Azure DevOps, Docker, Jira, Robot Framework, JUnit, Arduino

Concepts: Agile/Scrum, CI/CD, Mutation Testing, Test-Driven Development, RESTful Architecture, OOP, MVC